

Q1. What is the expansion of $(x + 3)^2$?

- (A) $x^2 + 6x + 9$
- (B) $x^2 - 6x + 9$
- (C) $x^2 + 3x - 9$
- (D) $x^2 - 3x - 9$
- (E) $x^2 + 9x + 3$

Q2. Which of the following is a perfect square trinomial?

- (A) $x^2 + 4x + 4$
- (B) $x^2 - 4x - 4$
- (C) $x^2 + 2x - 4$
- (D) $x^2 - 2x + 4$
- (E) $x^2 + 4x - 2$

Q3. What is the factored form of $x^2 + 10x + 25$?

- (A) $(x + 5)^2$
- (B) $(x - 5)^2$
- (C) $(x + 2)(x + 8)$
- (D) $(x - 2)(x - 8)$
- (E) $(x + 3)(x + 7)$

Q4. The gravitational force between two planets can be modeled by $F = \frac{Gm_1m_2}{r^2}$, where m_1 and m_2 are the masses of the planets and r is the distance between them. What is the expansion of $(r + 2)^2$ in this context?

- (A) $r^2 + 4r + 4$
- (B) $r^2 - 4r + 4$
- (C) $r^2 + 2r - 4$
- (D) $r^2 - 2r + 4$
- (E) $r^2 + 4r - 2$

Q5. A lab experiment involves the expansion of $(x - 4)^2$. What is the result?

- (A) $x^2 - 8x + 16$
- (B) $x^2 + 8x + 16$
- (C) $x^2 - 4x - 16$
- (D) $x^2 + 4x - 16$
- (E) $x^2 - 2x + 8$

Q6. A spacecraft's trajectory can be modeled using the equation $d^2 = (v_0t + \frac{1}{2}at^2)^2$. What is the expansion of $(v_0t + \frac{1}{2}at^2)^2$?

- (A) $v_0^2t^2 + v_0at^3 + \frac{1}{4}a^2t^4$
- (B) $v_0^2t^2 - v_0at^3 + \frac{1}{4}a^2t^4$
- (C) $v_0^2t^2 + 2v_0at^3 + a^2t^4$
- (D) $v_0^2t^2 - 2v_0at^3 + a^2t^4$
- (E) $v_0^2t^2 + v_0at^3 - \frac{1}{4}a^2t^4$

Q7. The expansion of $(x + 2y)^2$ is

- (A) $x^2 + 4xy + 4y^2$
- (B) $x^2 - 4xy + 4y^2$
- (C) $x^2 + 2xy - 4y^2$
- (D) $x^2 - 2xy - 4y^2$
- (E) $x^2 + 4xy - 2y^2$

Q8. Which of the following is not a perfect square trinomial?

- (A) $x^2 + 6x + 9$
- (B) $x^2 - 6x + 9$
- (C) $x^2 + 2x - 4$
- (D) $x^2 - 2x + 4$
- (E) $x^2 + 4x + 4$

Open Ended Questions (FRQ)

Q9. The distance between two galaxies can be modeled by $d = (v_0t + \frac{1}{2}at^2)$. Expand $(v_0t + \frac{1}{2}at^2)^2$ and show your work. What is the significance of the expansion in this context?

Q10. A satellite's position can be determined using the equation $x^2 + 10x + 25 = (x + 5)^2$. Factor the left side of the equation and compare it to the right side. What does this tell you about the satellite's position? Use math to support your answer.

Chef's Tips

- Tip 1: Encourage students to use diagrams and visual aids to help with expansion and factoring.
- Tip 2: Remind students to check their work by plugging the factored form back into the original equation.
- Tip 3: Emphasize the importance of using the correct formula for expanding special patterns, such as $(a + b)^2 = a^2 + 2ab + b^2$